Table S5. Exploratory B: case-level site balanced accuracy per representation, analysis, probe and evaluation set, with the generated judgement statements.

Probe	Set	Cases	Steps	BA (95% interval)	One-sided 95% upper bound	Exceeds 0.35	Judgement statements (pairs ending at this set)
FM-P, transport							
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S1 and S2
linear	S3	158	11850	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
MLP	S1	278	104662	0.957 (0.927 to 0.982)	0.979	yes	—
MLP	S2	158	70763	0.969 (0.936 to 0.994)	0.991	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.884 (0.827 to 0.933)	0.926	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
FM-P, within-set refit							
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.231 (0.182 to 0.282)	0.273	no	the judgement does not change between S1 and S2
linear	S3	158	11850	0.416 (0.352 to 0.481)	0.472	yes	the judgement changes between S2 and S3 the judgement changes between S1 and S3
MLP	S1	278	104662	0.957 (0.927 to 0.982)	0.979	yes	—
MLP	S2	158	70763	0.948 (0.907 to 0.982)	0.982	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.702 (0.640 to 0.763)	0.755	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
FM-S, transport							

Probe	Set	Cases	Steps	BA (95% interval)	One-sided 95% upper bound	Exceeds 0.35	Judgement statements (pairs ending at this set)
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S1 and S2
linear	S3	158	11850	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
MLP	S1	278	104662	0.495 (0.456 to 0.535)	0.531	yes	—
MLP	S2	158	70763	0.479 (0.433 to 0.523)	0.517	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.501 (0.443 to 0.562)	0.550	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
FM-S, within-set refit							
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.239 (0.188 to 0.290)	0.282	no	the judgement does not change between S1 and S2
linear	S3	158	11850	0.393 (0.346 to 0.444)	0.435	yes	the judgement changes between S2 and S3 the judgement changes between S1 and S3
MLP	S1	278	104662	0.495 (0.456 to 0.535)	0.531	yes	—
MLP	S2	158	70763	0.520 (0.475 to 0.566)	0.559	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.428 (0.368 to 0.489)	0.476	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
REF, transport							
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S1 and S2

Probe	Set	Cases	Steps	BA (95% interval)	One-sided 95% upper bound	Exceeds 0.35	Judgement statements (pairs ending at this set)
linear	S3	158	11850	0.250 (0.250 to 0.250)	0.250	no	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
MLP	S1	278	104662	0.863 (0.818 to 0.904)	0.900	yes	—
MLP	S2	158	70763	0.875 (0.822 to 0.924)	0.916	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.788 (0.724 to 0.848)	0.839	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3
REF, within-set refit							
linear	S1	278	104662	0.250 (0.250 to 0.250)	0.250	no	—
linear	S2	158	70763	0.208 (0.158 to 0.257)	0.251	no	the judgement does not change between S1 and S2
linear	S3	158	11850	0.448 (0.387 to 0.518)	0.505	yes	the judgement changes between S2 and S3 the judgement changes between S1 and S3
MLP	S1	278	104662	0.863 (0.818 to 0.904)	0.900	yes	—
MLP	S2	158	70763	0.871 (0.815 to 0.924)	0.918	yes	the judgement does not change between S1 and S2
MLP	S3	158	11850	0.620 (0.543 to 0.687)	0.680	yes	the judgement does not change between S2 and S3 the judgement does not change between S1 and S3

Exploratory (outside the frozen protocol; plan registered on OSF, osf.io/g9ymw, after the step-7 verdicts had been generated); no verdict. Transport: the probe fitted as in E3 on S1 and evaluated on the held-out cases of each set; within-set refit: the probe refitted within each set; the two are separate analyses and are never pooled or contrasted. Sets: S1, 278 cases, all steps; S2, 158 primary cases, all steps; S3, 158 primary cases, analysis region (75 steps per case). Steps: step rows evaluated.

BA, case-level site balanced accuracy (four centres; chance 0.25) after linear site removal; 95% interval and one-sided 95% upper bound: case-level bootstrap stratified by site (2,000 resamples). The boundary 0.35 applies to the upper bound. Judgement statements as generated by the analysis (b_judgements.csv) for the pairs S1→S2, S2→S3 and S1→S3, listed at the later set.

Ceiling (fact base §9): B locates where the judgement changes; it does not identify why. The subsampling changes centre balance, case set and steps per case at the same time. MLP, two-layer multilayer perceptron.